Sahyog Care Companion · Architecture pack

> **Written for the MVP.** This pack describes the MVP's code at `s7-accepted`. This final build has the same architecture, plus the fixes its reviews made (listed in `AI-COLLABORATION.md`): for example, a stricter grounding of medicines and doses in W3, a stricter wrong-patient check in W2 ingest, fenced ingest workers, a desk-move lock, and wider red-flag and wording rules. Where a detail below differs, the code and the log are right. **Numbers in sections 6 and 7 (recall, scan-read error rate, triage speed) are the MVP's and weren't re-run in this build.** The PDF is the same text.

Version	1.0, 2026-11-02
Describes	The MVP's code at its tag `s7-accepted`. This repo's `main` adds the review fixes (see the note above). The tag exists only in the MVP repo
Required by	Engagement pack p48: context, container, entity model, three graphs and two sequences, all corrected from the pack
Timing	Written after the build. The engagement pack asks for version one on day 5, before graph code, so this doesn't meet that ordering

How to read this

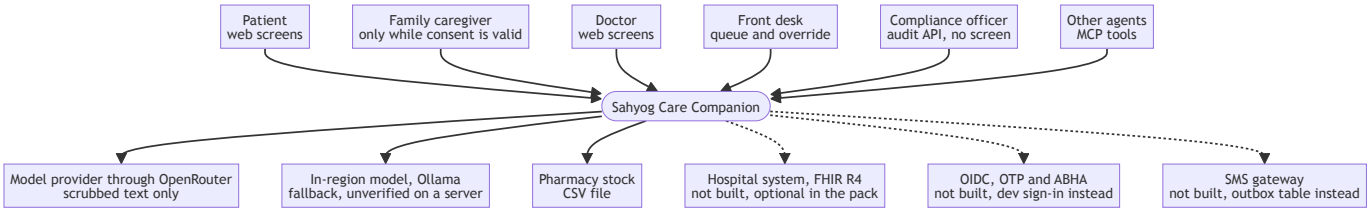
- **Every diagram shows what the code does today**, including where that differs from the engagement pack's design. Section 6 lists each difference and why it exists.
- **Dashed or labelled "not built" means it doesn't exist.** This is a demo on synthetic data, running on one Mac. ADR 0005 proposes how it would be deployed, and nothing there is deployed.
- **The design has one argument:** anything that can harm a patient is decided by code or by a named person, and the model only handles the fuzzy steps. A prompt can be argued with, but a SQL filter can't.

Colours in the graphs

Colour	Meaning
Grey	Plain code
Pink	Calls a model
Purple	Uses a tool or data service
Gold	Pauses for a person
Red	Ends in a hand-off to a person

1. Context view

Who uses the system and what it depends on.

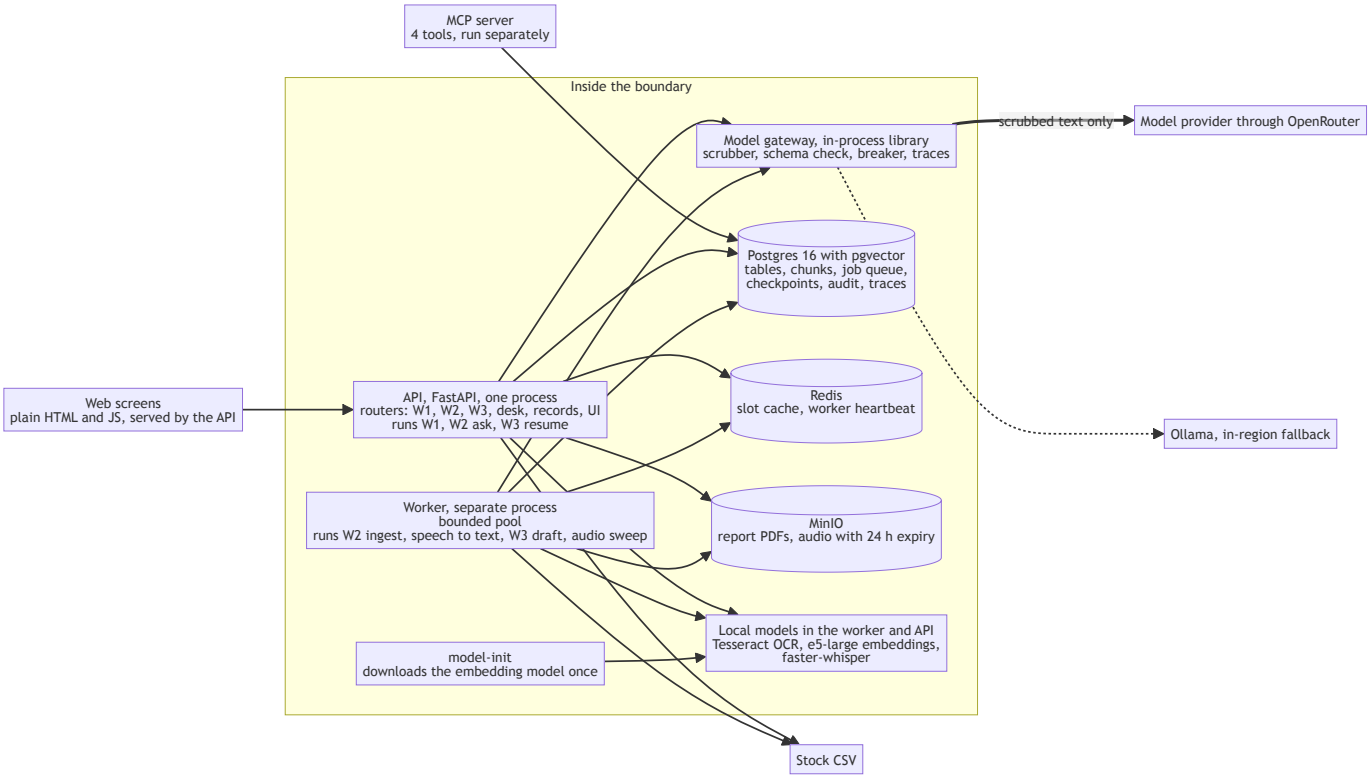


What each person is owed

Person	Fear	What the system shows them
Chief medical officer	AI advice harms a patient	Red-flag gate before any match, and refusal of diagnosis and dose questions (by the model's flag, with word checks as a backstop: they caught 6/25 unseen advice questions in this build, 0/7 in Hindi)
Patient	A stranger reads their reports	Patient filter in SQL, and citations on every answer
Doctor	Blamed for an AI note	Nothing is stored until they sign, and a named medicine is never invented
Front desk	Being replaced	Hand-offs come to their queue, and they can override any assistant booking
Compliance officer	A data-protection breach	Consent rows, and an audit event for each record read
Pharmacy	Software substituting medicines	Stock flags only, with the choice left to the doctor

2. Container view

Everything inside the dashed box runs on one machine under Docker Compose. Only scrubbed text leaves it.



Key choices

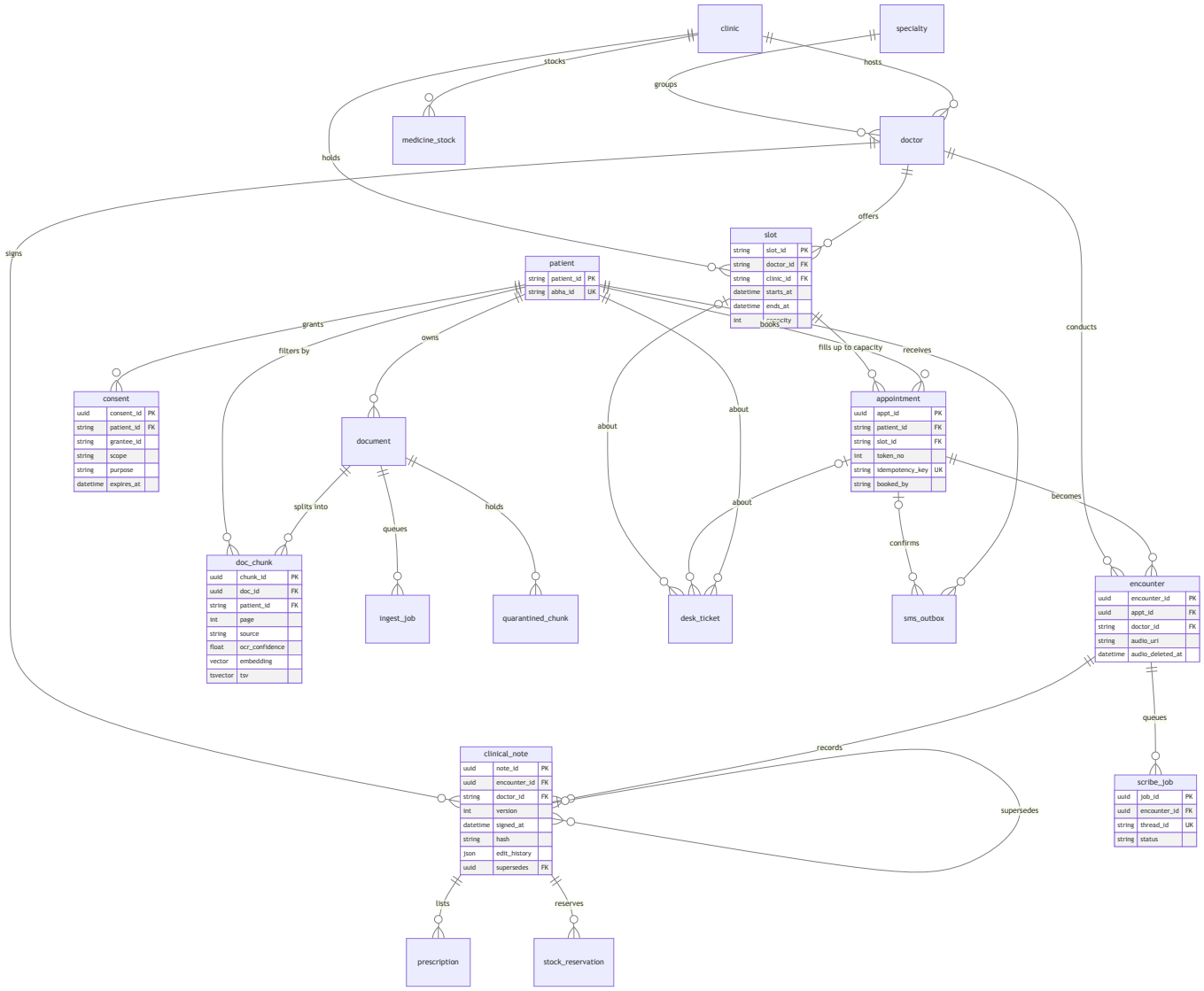
Choice	Why	What it costs
One API plus one worker, instead of five services	A two-week build, and modules keep the seams (`sahyog/scheduling`, `records`, `scribe`, `safety`)	They can't yet scale or deploy apart
Separate worker	OCR and embedding can't starve triage	One more process to watch, so `/health` reports its heartbeat
Postgres as the queue	`FOR UPDATE SKIP LOCKED` keeps the job and its document in one transaction	Held in the Mac test, but unproven on real infrastructure
Graph runs resume from checkpoints	Any API replica could resume any thread	Tested with one process only
One way out for clinical text	Every model call goes through the gateway, which scrubs first	It's a code-level rule. ADR 0005 adds a network allowlist, which isn't built

Guardrails on every model call (`sahyog/llm/gateway.py`)

- Scrubs identifiers before text leaves, and restores placeholders only for the caller.
- Rejects any reply that doesn't match the schema.
- Applies a timeout, and tries the primary route twice before the fallback.
- A circuit breaker gives a provider no calls for 30 seconds after 5 failures in a row. It's per process.
- Writes one `trace_span` row with tokens, cost and latency, and no prompt text.
- When the model fails, the run goes to the human desk.

3. Entity model

There are 23 application tables, from `sahyog/models.py`. LangGraph's checkpoint tables sit beside them. Only key columns are shown.



Tables with no foreign keys, by design

Table	Purpose
`audit_event`	One row per record read or denied action. It names the record by ID so it can outlive the record
`trace_span`	One row per model call
`idempotency_record`	Stored replies, so a retried request gets the first answer
`embedding_cache`	Embeddings by content hash

Rules the schema carries

- **A slot is a block of time with a capacity, not one appointment.** Patients get a token number and a time block, never an exact minute. A unique constraint on slot and token prevents two patients getting one token, and a database trigger rejects a token above the block's capacity.

- **The audit log is append-only, and a signed note can't change.** Both are enforced by database triggers, so the rule holds even for code that bypasses the API.
- **`doc_chunk` repeats `patient_id`** so the search filter doesn't depend on a join.
- **`prescription.medicine_id` isn't a foreign key,** so a doctor can prescribe something off the stock list, stored as `UNLISTED`.
- **`consent.grantee_id` isn't a foreign key,** because there's no caregiver table yet.

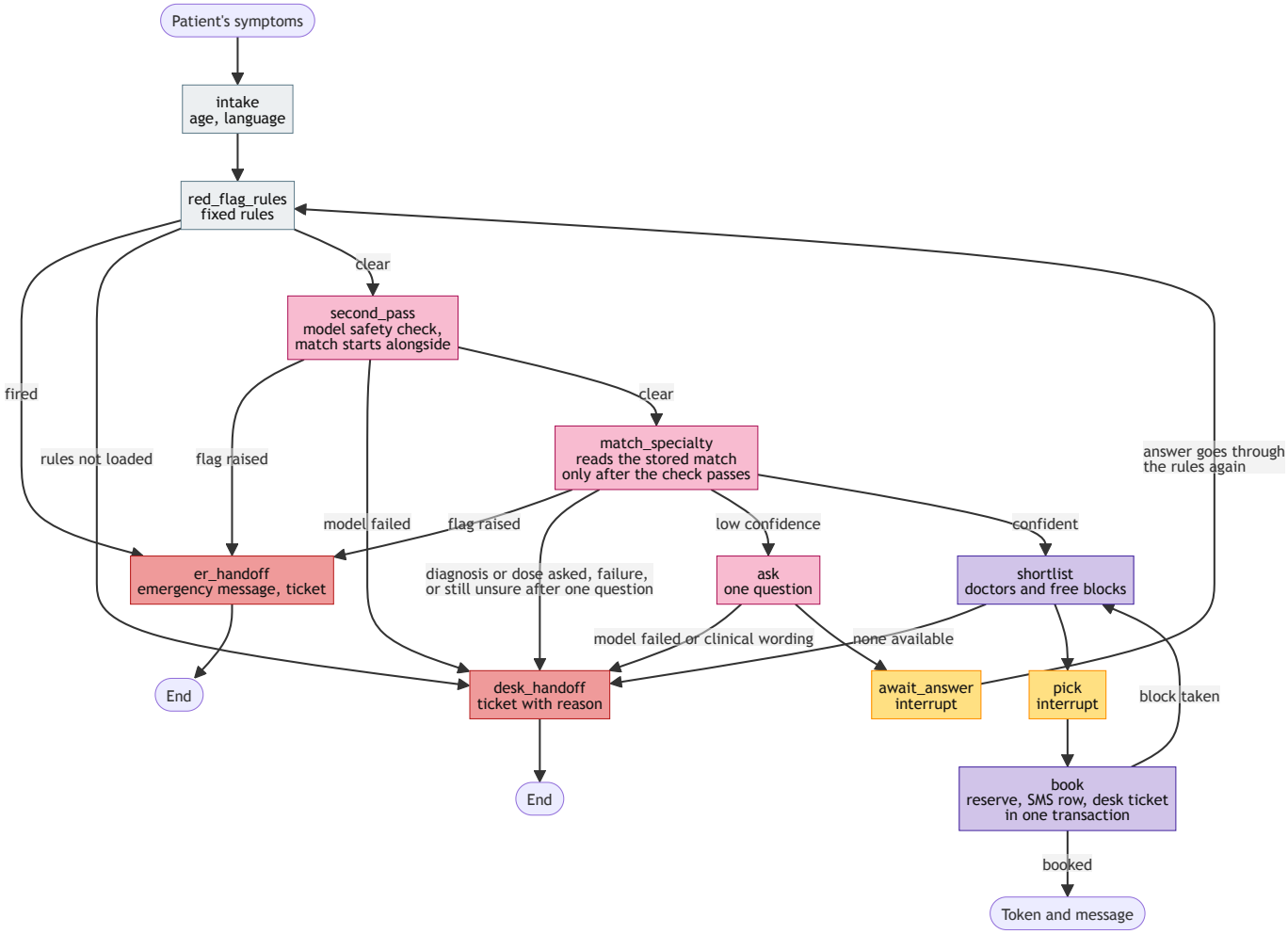
Gap: `docs/data-contracts/` has no contract for `scribe_job`, `stock_reservation` or `trace_span`.

4. The three workflow graphs

W2 is two compiled graphs, so there are four in all. Each has its code in `sahyog/graphs/`.

W1 · Find my doctor (`w1.py`)

Keeps state in the Postgres checkpointer and pauses twice for the patient.

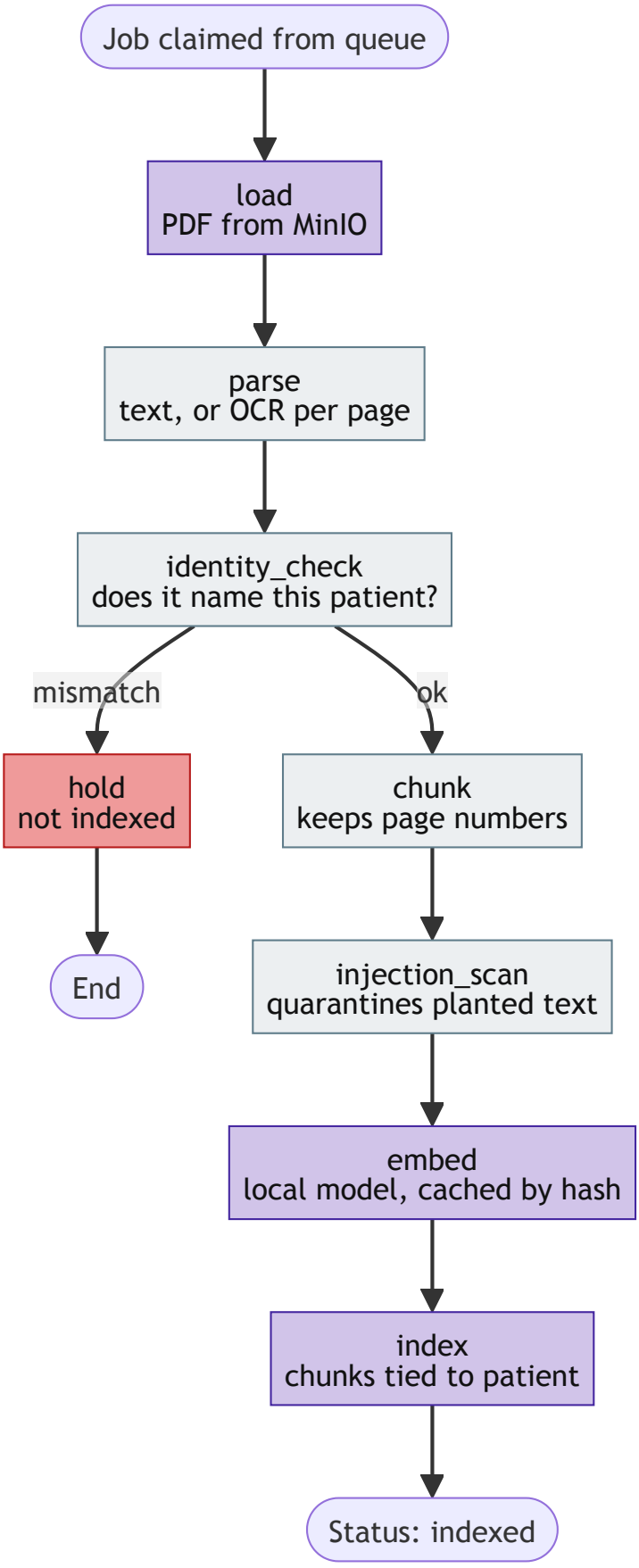


Why it's built this way

- **The safety check always comes first.** The match runs alongside it only to meet the 4 second target, and its result isn't read until the check passes.
- **A clarifying answer goes back through the rules,** because the answer itself may describe an emergency.
- **A gate with no rules fails closed.** If the rule list didn't load, the patient goes to the desk.
- **The patient can only pick a block they were offered.** The API checks this before resuming.

W2 · My records, ingest (`w2_ingest.py`)

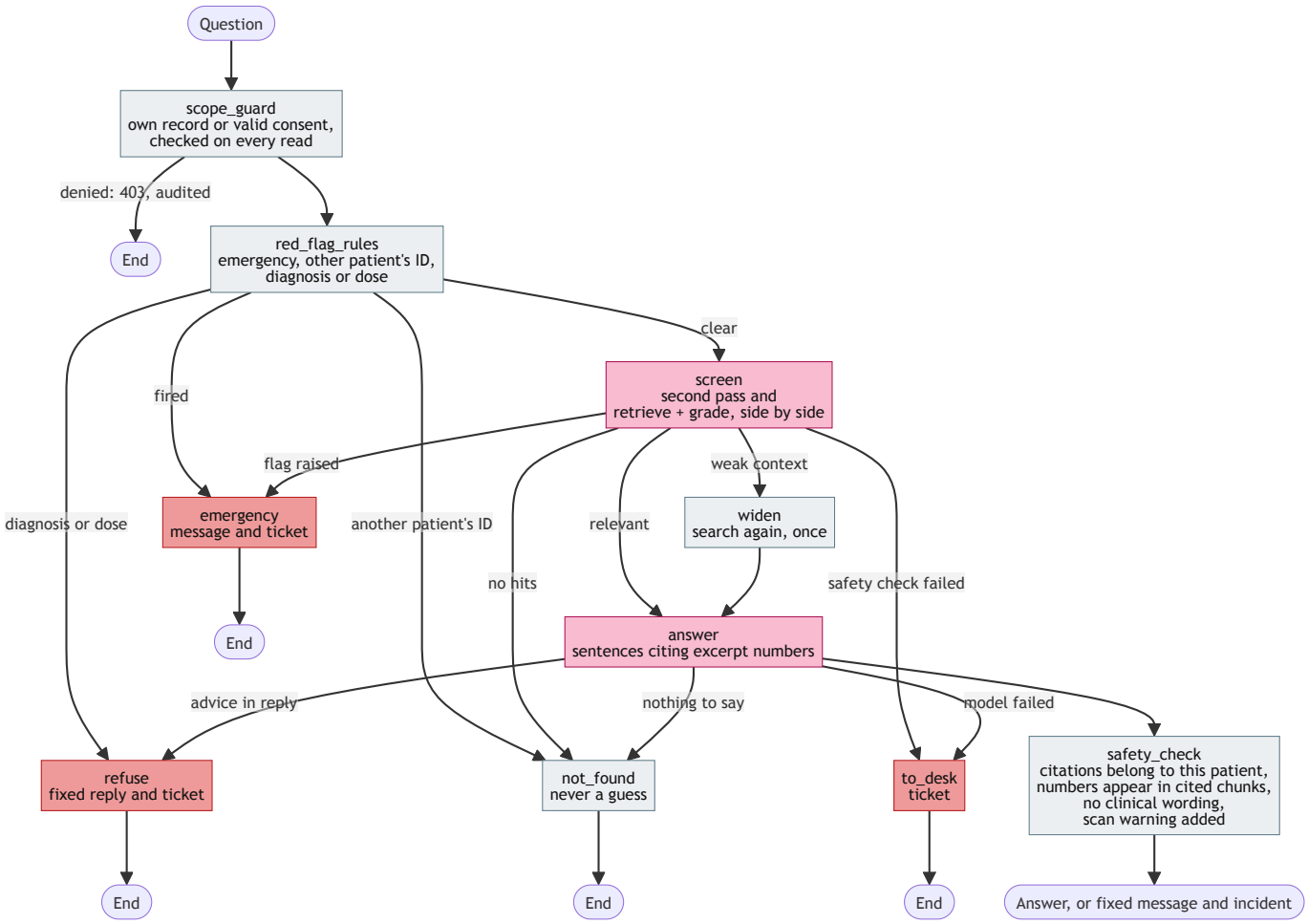
Runs on the worker. The `ingest_job` table holds its progress, so it has no checkpointer.



No model outside the boundary is called here. OCR and embeddings are both local.

W2 · My records, ask (`w2_ask.py`)

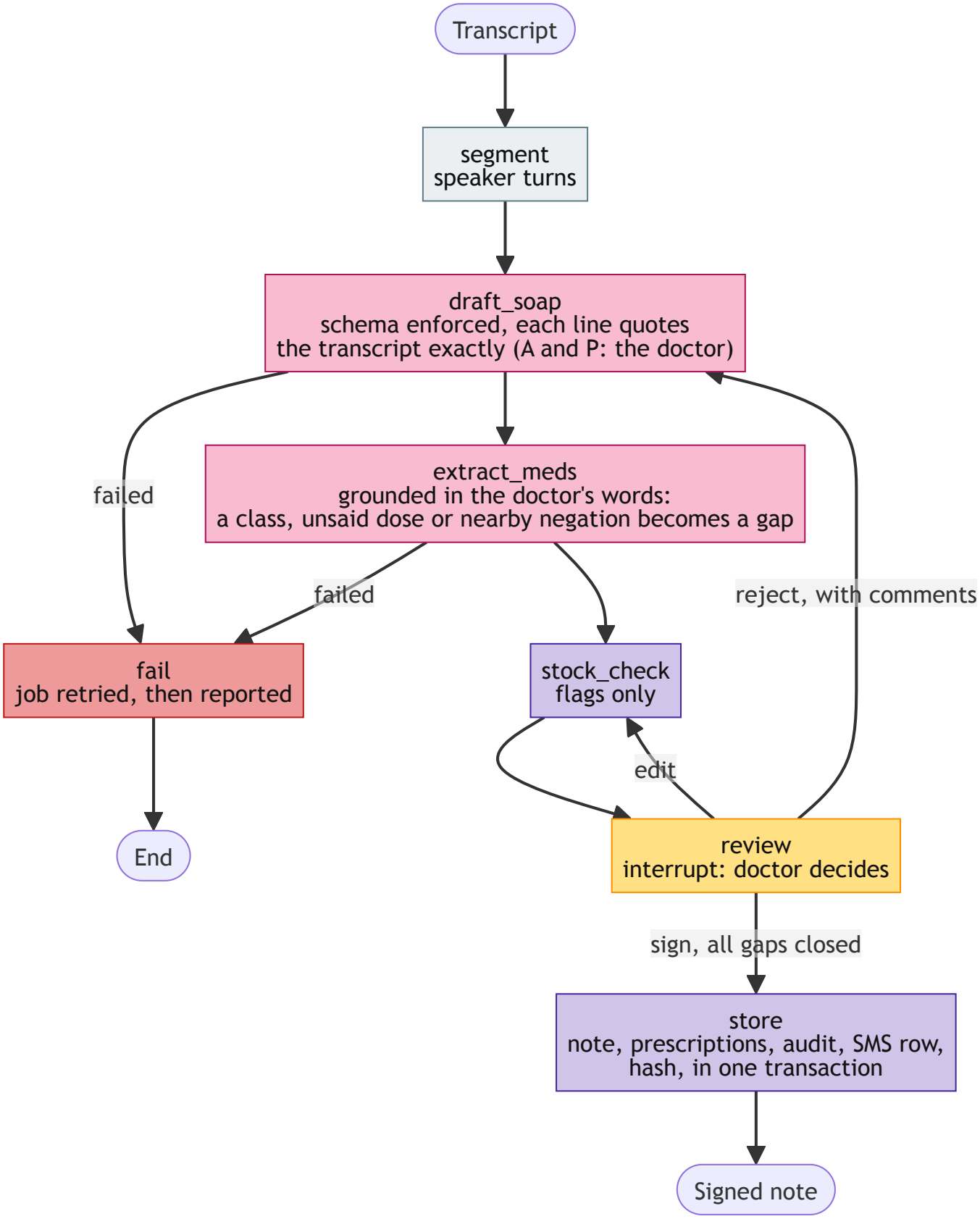
Runs on the request and streams only after the safety check, so a blocked answer is never partly shown.



Why a patient can't see another patient's data: the filter is in the SQL (`records/search.py`), and its `patient_id` is checked against the login (`can_read` in the scope guard) before any search, never taken from the model. A poisoned PDF can change what the model says, but it can't widen a `WHERE` clause. The red-team set tests this end to end.

W3 · Consultation scribe (`w3.py`)

The worker runs it up to the doctor's review. The API resumes it from the checkpoint when the doctor decides.



Why it's built this way

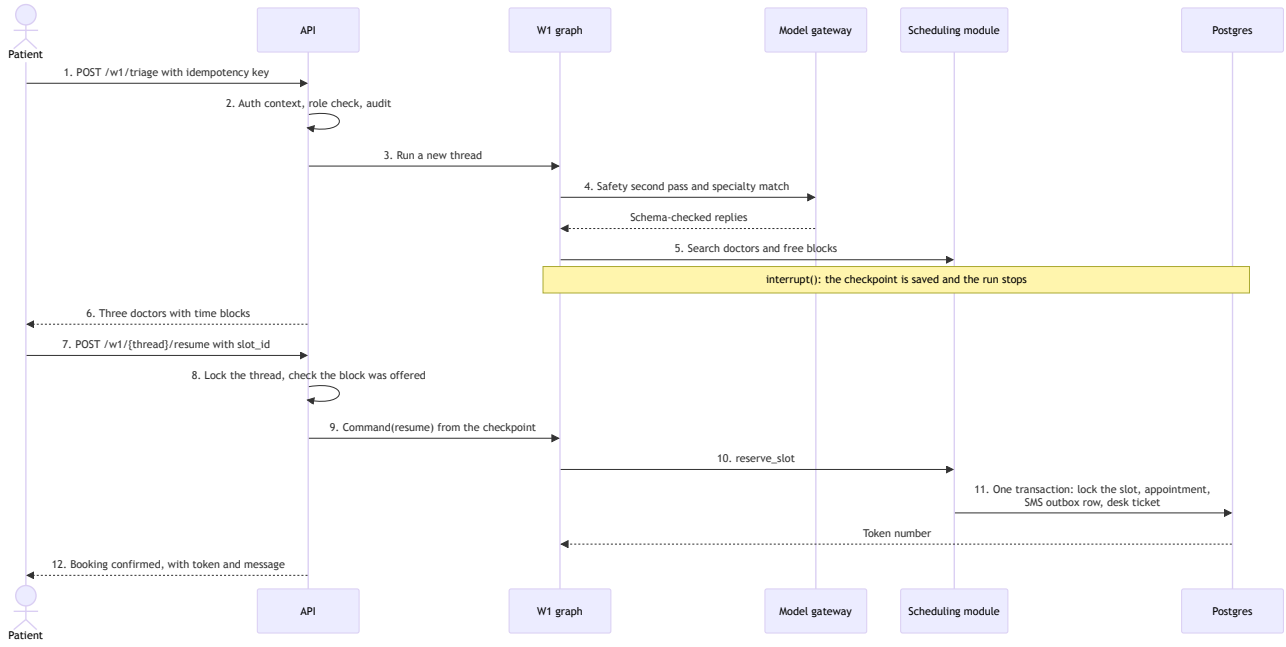
- `store` **is the only node that writes a clinical act,** and it can only run after a signature.
- **The API refuses to sign while a gap is open.** The model can't fill a molecule the doctor didn't say. If the doctor said "a statin", the doctor has to name it.

- **Grounding trusts the transcript.** It can't catch a speech-to-text error, such as "Known diabetic" heard as "NON diabetic". The review screen shows the transcript and a warning, but nothing forces the doctor to read it.

5. The two sequences

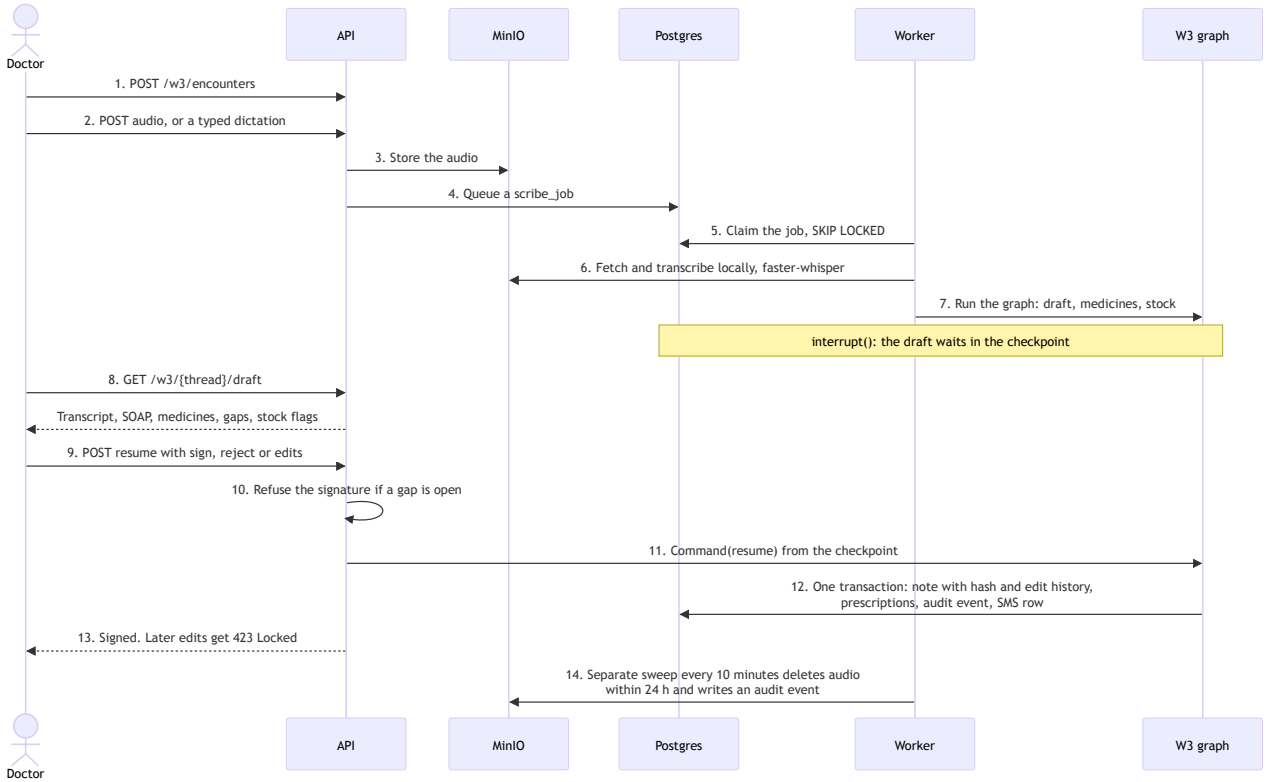
Booking a slot through W1

The API holds no conversation in memory. Between messages 6 and 7, the run exists only as a checkpoint in Postgres.



A retry with the same key returns the first booking. A retry can't send a second SMS, because the SMS row is written only for a new booking.

A consultation becomes a signed note



The bucket's expiry rule is the backstop for step 14. The sweep exists so that "who deleted it, and when" has an answer.

6. Corrections from the engagement pack

Each row is a place where this build departs from the pack's design.

Area	The pack says	What's built	Why, and what it means
Gateway	API gateway with OIDC, rate limits, routing and audit headers	None. FastAPI is reached directly	**Rate limiting isn't built,** so the pack's slot-hoarding test (1,000 bookings get twenty) would fail as written. A patient can hold only one place per block
Identity	OIDC with OTP, and ABHA as an extension	Local signed tokens and a dev sign-in	Must stay off in any shared environment
Services	Agent, scheduling, records, scribe, audit and consent services	One API and one worker, with modules	Faster to build, and the module seams remain
LLM route	LLM API	OpenRouter, a further third party	The user's choice (ADR 0004). India-only processing isn't provided on this route, and some other people's names can leak
Hospital system	FHIR R4 reader	Not built	Optional in the pack. Demographics come from seeded tables
Stock	One interface with CSV and API adapters	The interface and the CSV adapter	The API adapter waits on the client's API
Slots	Free-or-taken slots, with capacity as the fallback	Blocks with capacity and tokens	Nobody can predict consultation length, so exact minutes aren't promised
W1	Rules, match, shortlist, options, pick, book	Adds a model second pass, and hand-offs from every step	Rules alone caught about half of unseen phrasing. The second pass narrowed that gap on the MVP's probes, one of which wasn't blind. The independent check hasn't run
W2 ingest	Tags PHI with Presidio while ingesting	Adds an identity check and injection scan, and scrubs at the gateway instead	Chunks stay as printed, inside the boundary, and scrubbing happens only when text leaves
W2 ingest	Parallel ingest with the Send API	A bounded worker pool on the queue, no Send API	**Misses an Architect-tier item**
W2 ask	Scope, retrieve, grade, answer, safety	Adds a red-flag check, number check and scan warning	A patient may mention an emergency while asking about a report, and about 1 in 200 scan-read numbers is wrong (MVP measurement)
W3	STT is a separate service, and a timeout parks the draft	STT runs in the worker, and the draft simply waits with no timeout	Simpler, with no audio leaving the boundary. A forgotten draft stays open
Booking sequence	Sends SMS through a gateway	Writes an outbox row in the booking transaction	No real gateway exists, so no SMS is actually delivered
Memory	Long-term store keyed by patient	Not built	Language comes from each request, and nothing is remembered between threads
Thread expiry	Checkpoints purged after 30 days	A manual command, `make purge-threads`	Nothing runs it on a schedule, so thread state stays until someone does

Area	The pack says	What's built	Why, and what it means
Tools	Timeouts and backoff on every tool	Timeouts and the breaker on model calls, and retries on database nodes	Other tools have no timeout of their own
Traces	Langfuse, with spans per node	A Postgres table, model calls only	Langfuse was too heavy for the machine (ADR 0004)
Judge	Calibrated, using ragas	Uncalibrated, using the published method through the gateway	Ragas would bypass the scrubber. Calibration needs 15 more human grades

7. What this architecture hasn't proved

- **Scale:** it has run on one Mac with one API process. Triage narrowly missed its 4 second p95 at the target rate, at 4.28 s (MVP load test, not re-run).
- **Emergencies:** the independent test of emergency phrasing has never run. This is the top risk in the final report.
- **Clinical safety:** no clinician has reviewed the rules, prompts or messages.
- **Real documents:** scans and audio have all been synthetic.
- **Recovery:** no backups exist, and restore is untested.
- **MCP:** its tools were called directly and listed, but not from a real MCP client.

Related documents

- Decisions: ADRs 0001–0005
- Data rules: data contracts and red-flag rules
- Results: final report
- Walkthrough: demo script